

NAYEEM HABIB

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Education

NIT , TIRUCHIRAPPALLI

Aug 2019 - Jun 2023

B.Tech - Electrical and Electronics Engineering

CGPA: 7.2/10

Relevant Coursework: Data Structures and Algorithms, Digital Electronics and Analog Electronics etc.

Skills

Languages: C/C++, Java, JavaScript, HTML, CSS, Python, SQL and Verilog.

Technologies & Tools: VS Code, Xilinx- Vivado, LtSpice, Matlab, Power Bi, Microsoft Excel, Git and GitHub.

Work Experience

UPSC Civil Services Preparation.

Jun 2023 - Dec 2025

Career Break – UPSC Preparation

- Prepared for the UPSC exam with **Electrical Engineering (EEE)** as my optional subject.
- Strengthened core concepts in **Power Systems, Control Systems, Machines, and Analog and Digital Electronics.**
- Developed strong analytical, time management, and self-discipline skills.

IIT Madras, Chennai.

May 2022 - Jul 2022

Research Internship

- Implemented the Particle Swarm Optimization algorithm in Verilog, achieving a **processing speed of 30** percent compared to previous implementations and yielding insights for future optimization projects.
- Analyze three Verilog modules to display a **specific 16-bit floating-point number on the FPGA board.**
- Used **Matlab, C++, FPGA Board for Project.**

NIT Tiruchirapalli, Tamil Nadu.

May 2021 - Jul 2021

Research Internship

- Developed a MultiFace LIPSYN Simulation Model using deep learning.
- Used Speaker Diarization, Face Detection and Recognition, Lip Synchronization Smoothing and Super Resolution using **GEPGAN.**
- Used DeepFilterNet, CNN, TensorFlow, Keras, PyTorch and NumPy.

Project Work

- **RLC filter Circuit Design and Analysis(March 2023 - April 2023):** Developed a simulation model of an RLC filter circuit using Python, achieving an **accuracy of up to 98 Percent.** Analyzed over **100 configurations** of RLC circuits to identify the optimal configuration. Used **Python, NumPy, Matplotlib and SciPy.**
- **Digital Signal Sampling and Analysis (June 2022 - July 2022):** Simulated DSP concepts such as **sampling, aliasing, DFT, and signal** reconstruction using MATLAB. Analyzed spectral resolution, frequency spectrum, and interpolation techniques for discrete-time signals. Validated **Nyquist-Shannon sampling** theory through practical signal processing experiments. Used **Python, NumPy, Matplotlib and SciPy.**
- **Circuit Delay Prediction for CMOS Timing Analysis(Jan 2022 - Feb 2022):** Developed a Simulation model of traffic light control system using AI With **accuracy to 99 Percent.** Utilized computer vision techniques to determine time limits on roads. Used **Python, Image Processing, Object Detection and TensorFlow.**
- **Face Recognition Biometric system(Nov 2021 - Dec 2021):** Developed a simulation model of face recognition biometric system. Used **Python, Image Processing, Image Recognition, OpenCV, SciPy, SciKit.**
- **VEHICLE TRACKING SYSTEM(Aug 2021- Sep 2021):** Developed a GPS tracker using Arduino to transmit location data to mobile devices via a GSM module; successfully tested on more then **10 devices**, with reliable performance over a **distance of up to 1 km.** Used Arduino UNO, GSM Module, GPS Module, Power Supply,Connecting Wires.

Responsibilities

- Team Leader, Analog Domain, **Tronix 3D Aeromodelling** Club, NIT Trichy.
- President for organizing Current Events, NIT Trichy.
- Workshop manager for EEE Association's NIT Trichy.
- Class representative for **3rd and 4th year**, NIT Trichy.

Certificates

- Geo-Data Sharing and **Cyber Security(ISRO).**
- Supervised Machine Learning: Regression and Classification.
- Advanced Learning Algorithms
- **Certificate of Excellence** in Football for inter school performance.